Learning Illumination to Reduce Structural Ambiguity in 3D Fluorescence Microscopy

Anonymous CVPR submission

Abstract

A computational microscope returns one volume, but its measurements may admit many. A reconstruction prior favours one explanation and can introduce connections that the images do not establish. We learn illumination to reduce this structural ambiguity in fixed-focus three-dimensional fluorescence microscopy. Scanning the incident angle through a fixed, defocused phase plate produces depth-dependent excitation patterns while the specimen and detection focus remain stationary. We train the plate to decorrelate measurement signatures across depth, without specimen examples or a reconstruction solver in the training loop. We assess both reconstruction quality and measurement-supported connectivity. A connectivity query is certified when every volume consistent with the images and stated error model agrees under a specified intensity threshold and connectivity definition. Two consistent volumes with opposite connectivity demonstrate ambiguity. On reduced-field synthetic volumes, certified connections increase from 3 to 12 of 48 connected queries, with none lost. On simulated measurements of mouse pyramidal neuron volumes, it reduces mean local depth error from 0.43 to 0.23 μm at eight-fold measurement compression with a pretrained prior. Neuron-trained denoisers further improve recovery, but do not resolve the tested in-focus amplitude encoder's depth ambiguity. This approach enables learned illumination to be evaluated by both reconstruction accuracy and the structural conclusions supported by its measurements.

1. Introduction

Fixed-focus three-dimensional fluorescence microscopy encodes depth into planar measurements and recovers the volume computationally. When different volumes explain the same images, a reconstruction prior favours one explanation. The resulting volume may contain plausible structures that the measurements alone do not establish [2, 6]. For neuronal imaging, this distinction matters when assessing whether fluorescent branches are connected. A reconstruction may show a continuous path even when a disconnected alternative remains consistent with the images.

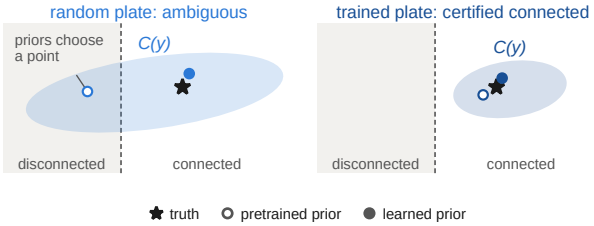


Figure 1. **Reconstruction alone does not certify connectivity** (schematic). Left: with the random plate, the measurement-consistency set $\mathcal{C}(\mathbf{y})$ contains both connected and disconnected volumes. A prior selects one reconstruction, but the alternative remains admissible. Right: an illustrative outcome of plate training, where every volume consistent with the new measurements and stated error model is connected, certifying the query. Reconstruction selects one explanation; certification establishes agreement across all admissible explanations.

Learning can improve both optical encoding and reconstruction. End-to-end methods optimise the optics together with a reconstruction model [8, 19, 26], while measurement-based approaches optimise optical responses or information capture independently of a decoder [21, 25, 30]. Improved reconstruction accuracy, however, does not necessarily establish whether a particular connection is supported by the measurements. We therefore ask whether learning the illumination can reduce structural ambiguity, so that all volumes consistent with the images and stated error model agree on connectivity (Fig. 1).

We study this on a fixed-focus fluorescence microscope that encodes depth in the excitation (Fig. 2). A plane wave is scanned in angle through one fixed phase plate placed a short distance before an intermediate image plane; a high-NA relay maps the field into the specimen; a low-NA camera at fixed focus records one image per angle. A tilt translates the diffracted pattern by an amount that grows with propagation distance, so each depth sees its own sequence of excitation values without axial specimen scanning or detection refocusing. A learnable phase map controls the depth encoding, while a voxel-integrated vectorial model allows us to evaluate candidate volumes against the measurements

under explicit object and error assumptions.

Our contributions are:

1. **Learning illumination for depth discrimination.** We develop a specimen-agnostic objective for optimising a fixed, defocused phase plate by decorrelating measurement signatures across depth. Training incorporates the illumination and detection model without a reconstruction solver.
2. **Evaluating measurement-supported connectivity.** We apply connectivity certification to evaluate learned illumination independently of reconstruction quality. In reduced-field simulations, the trained plate increases certified queries from 24 to 33 of 72, retaining every query certified with the random plate. Connected and disconnected volumes consistent with the same measurements demonstrate the remaining ambiguity under the stated error model.
3. **Separating illumination and prior gains.** Using a voxel-integrated vectorial forward model and matched excitation dose, we compare optical and prior improvements separately. At eightfold measurement compression, plate training reduces mean local depth error from 0.43 to 0.23 μm with a pretrained prior. Neuron-trained denoisers improve recovery with the phase plate, but do not resolve the tested in-focus amplitude encoder's depth ambiguity.

2. Related work

Depth encoding in fluorescence microscopy. Light-field microscopy, DiffuserCam and Miniscope3D encode volumetric information in the detection response [1, 4, 30]. Excitation coding offers a complementary route. Yeh *et al*. [31] reconstruct three-dimensional fluorescence using translated speckle illumination and axial sample scanning, while Pascucci *et al*. [20] exploit axial speckle decorrelation to recover volumes from a two-dimensional scan in both linear and saturated excitation regimes. These studies establish speckle-based depth encoding. Our work examines a learned excitation encoder operated with a stationary specimen and fixed detection focus.

Angular scanning and depth-dependent shifts. Yilmaz *et al*. [32] vary the incident angle through a fixed scattering layer for two-dimensional fluorescence recovery. Jiang *et al*. [11] use tilted pattern projection to produce depth-dependent lateral shifts, enabling volumetric imaging without axial scanning. Our acquisition builds on these mechanisms. Scanning the incident angle through a fixed, defocused phase plate produces excitation sequences that vary with depth. We study this arrangement in simulations of linear fluorescence with conventional fixed-focus camera detection.

Learning optical encoders. Optical optimisation need not involve a reconstruction network. Shechtman *et al*. [25] design detection-side phase masks using Fisher information, while Miniscope3D [30] penalises cross-coherence between depth-dependent point-spread functions. Coherence minimisation also has a broader history in sensing design [5]. Reconstruction-driven alternatives include learned coded illumination [13], differentiable fluorescence microscopy [8] and joint phase-mask/localisation-network design [19]. More recently, IDEAL [21] and IDEAL-IO [17] optimise estimated mutual information over measurement distributions independently of a reconstruction decoder. Our objective instead reduces centred correlations between voxel measurement signatures at different depths. Each signature collects detector-integrated measurements across illumination angles, and training requires no specimen examples. We then evaluate whether the learned illumination improves reconstruction accuracy and makes additional connectivity queries certifiable under the stated model.

Structural ambiguity and certification. Set-membership estimation characterises solutions compatible with bounded errors [18]. Haldar [7] derives entrywise bounds over nearly data-consistent solutions, while hallucination analyses examine prior-induced errors and ambiguity imposed by the forward model [2, 10]. Bayesian approaches also assess structural hypotheses: BUQO [22] tests whether credible regions admit structure-absent alternatives, and QuantifAI [14] incorporates learned convex priors. Our analysis asks whether all volumes satisfying intensity bounds and calibrated measurement intervals agree on thresholded connectivity. Flow and separator formulations certify agreement or construct opposite-connectivity witnesses. Unlike methods that select a curvilinear network under structural constraints [27], this analysis tests whether alternatives remain admissible. We apply it on reduced fields to evaluate illumination training, independently of the learned reconstruction prior and conditional on the stated object and error models.

Forward modelling and reconstruction. Incoherent structured-illumination models combine excitation intensity with the emission response [31], while wave-optics models account for diffraction in volumetric imaging [4]. We integrate fine-grid optical responses over reconstruction voxels and compare total variation with plug-and-play recovery [28]. Our reconstruction adapts multi-slice fusion [15] to a three-dimensional ADMM framework, using lower-dimensional denoisers to regularise a shared volume. These established components support separate evaluation of illumination and prior gains; we do not propose a new reconstruction algorithm.

3. Angle-scanned phase-encoded 3D imaging

Acquisition. We vary the illumination direction across two angular coordinates while keeping the phase plate fixed (Fig. 2). For transverse position $\mathbf{r} = (x, y)$, the incident plane wave is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^\top \mathbf{r}}$, with $\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t)$, where α_t is the azimuth and θ_t parametrises the direction in the vertical plane containing that azimuth. The corresponding forward direction is $(\mathbf{s}_t, \sin\theta_t)$, and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for the simulated refractive index of one. Thus, the two coordinates control the orientation and signed magnitude of the transverse phase gradient. The scan contains 256 distinct directions with $0.5 \le \|\mathbf{s}_t\| \le \sqrt{3}/2$. Each incident field passes through the phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The NA-0.8 objective pupil filters the modulated field before a vectorial Richards–Wolf relay [16, 23] produces the specimen excitation $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$, assuming x-polarised illumination. Coordinates and distances are expressed in specimen-equivalent units. The camera records one fluorescence image per direction at a fixed detection focus. Acquisition is sequential, not single-shot.

Why the angle encodes depth. Propagation converts the plate's phase structure into spatially varying excitation. Scanning the incident angle shifts this pattern differently at different depths, giving each location a depth-dependent illumination sequence. An ideal scalar Fresnel model illustrates the mechanism: without pupil clipping, an angular change $\Delta\mathbf{s}$ displaces the pattern by $(D + z)\Delta\mathbf{s}$, where $D + z$ is the distance from the plate. Different depths therefore translate at different rates, while plate defocus positions the shear pivot outside the specimen. This relation provides intuition; our high-NA vectorial model captures the finite pupil and departures from pure translation.

Image formation. Let $X_0(\mathbf{r}, z)$ denote the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z') X_0(\mathbf{r}', z') I_t(\mathbf{r}', z')\, \mathrm{d}^2\mathbf{r}'\, \mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity point-spread function and Ω is the specimen volume. Pixel p, occupying region D_p, records the normalised sample $\hat{Y}^0_{t,p} = |D_p|^{-1} \int_{D_p} g_t(\mathbf{r})\, \mathrm{d}^2\mathbf{r}$. Fluorescence contributions therefore add incoherently before pixel averaging. Multiplication by the excitation pattern mixes specimen spatial frequencies before detection blur, so the camera need not resolve the illumination pattern itself. Depth discrimination depends on how these measurements vary across illumination angles.

Voxel-integrated operator. To capture excitation variations within reconstruction voxels, we evaluate the optics on a finer grid and integrate the measurement responses over each voxel. Let B be the fine-grid measurement operator, including emission propagation and pixel averaging, and let U replicate each reconstruction coefficient over its fine-grid cells. Then $\mathsf{A} = \mathsf{B}\mathsf{U}$ preserves the fine-grid response for every specimen represented in this piecewise-constant basis.

Sub-voxel specimen structure remains unrepresented. For a fine-grid specimen $\mathbf{x}_f$, its coarse representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, obtained by voxel averaging, satisfies

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{U}\mathsf{R}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Thus, integration retains within-voxel optical variation, while $\boldsymbol{\delta}_{\mathrm{disc}}$ captures specimen approximation error.

Linear, incoherent fluorescence enables efficient assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells, $(\mathbf{r}_i, z_i)$ is the location of cell i, and $g_{p,i}$ is its emission footprint at pixel p, including its integration weight. These footprints are computed once and reused across angles. After precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF\log F)$ for constructing columns through separate FFT-based forward simulations, with T patterns, P detector samples, F fine-grid cells and N reconstruction voxels.

Reconstruction. We combine least-squares data fitting with multi-slice denoising and sparsity through plug-and-play ADMM [15, 28]. The branches $\mathcal{J} = \{xy, yz, zx, \mathrm{sp}\}$ comprise three slice-denoising branches and one sparsity branch. For orientation o, P_o rearranges the shared volume into slices, which a pretrained 2D DnCNN $\mathcal{D}$ [33] denoises independently. The corresponding volume denoiser is $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$. Each iteration solves

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \sum_{j\in\mathcal{J}} \tfrac{\rho_j}{2}\|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2 \tag{4}$$

by conjugate gradients, followed by denoising $\mathbf{v}_o^{k+1} = \mathcal{D}_o(\mathbf{x}^{k+1} + \mathbf{d}_o^k)$, soft thresholding $\mathbf{v}_{\mathrm{sp}}^{k+1} = \mathcal{S}_{\lambda/\rho_{\mathrm{sp}}}(\mathbf{x}^{k+1} + \mathbf{d}_{\mathrm{sp}}^k)$, and dual updates $\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}$. We initialise from normalised back-projection and clip the final estimate to $[0, 1]$. These updates do not enforce the certification intervals; membership in $\mathcal{C}(\mathbf{y})$ (Eq. (6)) is checked separately when claimed. As a non-learned comparator, we use box-constrained least squares with isotropic 3D total variation [24], solved by ADMM [3]. Solver parameters are given in the supplement.

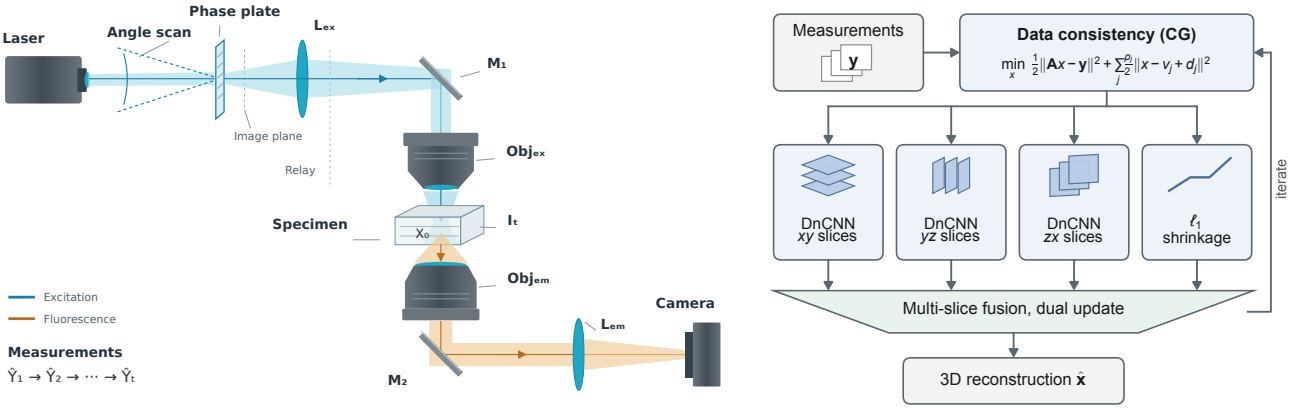


Figure 2. **Fixed-focus acquisition and reconstruction.** Left: a tilted plane wave illuminates a fixed phase plate placed 6.6 μm (specimen-space units) before an intermediate illumination image plane; an NA-0.8 vectorial relay maps the field into the specimen; a camera at a fixed detection focus records one image per illumination angle. Only the input angle changes: plate, specimen and focus are stationary, and nothing is scanned axially. Blue: excitation; orange: fluorescence. Right: multi-slice plug-and-play recovery alternates a conjugate-gradient data-consistency step with one pretrained 2D denoiser applied to the xy, yz and zx slices of a shared volume and an ℓ_1 shrinkage, fused by ADMM.

Learned prior. To adapt the reconstruction prior to neuronal structure, we evaluate two neuron-trained alternatives to the pretrained natural-image DnCNN: a 2D network shared across the three slice orientations, and a 3D network that replaces all three slice-denoising branches [12, 34]. Both are trained on neuron volumes from an animal excluded from testing, using Gaussian noise with $\sigma \sim \mathcal{U}(0, 0.25)$ and spatial and intensity augmentation. The data term, sparsity branch, iteration count and parameter-selection protocol remain unchanged. Network architectures and training details are given in the supplement.

Trained plate. We optimise the plate phase ϕ to reduce correlations between measurement signatures at different depths, with the aim of improving depth discrimination. For a voxel at lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ denote its centred, unit-norm operator column (Eq. (3)), collecting detector responses across illumination angles. We minimise

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r}} \sum_{z \neq z'} w_{zz'} \left(\mathbf{a}_{\mathbf{r},z}^{\top} \mathbf{a}_{\mathbf{r},z'}\right)^2, \tag{5}$$

where mirror-depth pairs about the detection focus receive weight two and other pairs receive weight one. Squaring penalises correlations of either sign; normalisation prevents uniform amplification of the signatures from reducing the loss.

Training differentiates through the excitation, vectorial relay and detection model, without specimen examples or a reconstruction solver. The objective is a surrogate for depth discrimination, rather than a guarantee of stable volume recovery. We assess its benefits through reconstruction and measurement-supported connectivity under matched excitation dose. Training details are given in the supplement.

4. What the measurements establish

A reconstruction is one volume that explains the images under a prior. For relational questions, such as whether a dendrite reaches a soma, we also ask whether *every* volume that explains the images agrees.

Consistency set and connectivity query. Given measurements $\mathbf{y} \in \mathbb{R}^M$ from k directions and per-row intervals $[\ell_i, h_i]$ that contain the noise-free measurement with stated probability, the measurement-consistency set is

$$\mathcal{C}(\mathbf{y}) = \{\mathbf{x} \in [0,1]^N : \ \ell_i \leq (\mathsf{A}\mathbf{x})_i \leq h_i, \ i = 1, \ldots, M\}, \tag{6}$$

where the box is the reconstruction's intensity normalisation. For calibrated Gaussian noise we use the Bonferroni box $y_i \pm z\sigma$ with $z = \Phi^{-1}(1 - \alpha/2M)$, $\alpha = 0.05$, so that $\mathbf{x}_{\mathrm{gt}} \in \mathcal{C}(\mathbf{y})$ with probability at least 0.95 under the noise model. In a query region with 6-adjacency, a voxel is active if $x_v \geq \tau$, and p, q are connected if an active path joins them; equivalently the bottleneck $b(\mathbf{x}) = \max_{P:p\to q} \min_{v\in P} x_v \geq \tau$. The query is *certified connected* if $\min_{\mathcal{C}} b \geq \tau$, *certified disconnected* if $\max_{\mathcal{C}} b < \tau$, and *ambiguous* otherwise. Ambiguity is proven by exhibiting two volumes in $\mathcal{C}(\mathbf{y})$ with opposite connectivity, both re-checked independently of the optimiser (unscaled residual, breadth-first search and union–find). Problems that a solver cannot decide within its budget are reported as undecided, never as ambiguous.

Optics change $\mathcal{C}$, priors choose in it. No prior enters Eq. (6): a reconstruction that fits the data is one member of $\mathcal{C}(\mathbf{y})$, and a better prior moves it within the set without changing what is certified. Only A and the noise level change

$\mathcal{C}$. The plate objective (Eq. (5)) acts on A in the direction that matters. For two voxels with unit-norm columns of correlation ρ in an ℓ_2 consistency ball of radius ε, the range of either voxel is $\pm\varepsilon/\sqrt{1-\rho^2}$, from the Schur complement of the 2×2 Gram matrix. Lowering the correlation between the columns of voxels at different depths therefore narrows their bounds. We use this as an intuition, not a theorem, since it ignores coupling among more than two voxels; Sec. 6.3 tests its size.

Computation. Per-voxel bounds $L_v = \min_{\mathcal{C}} x_v$, $U_v = \max_{\mathcal{C}} x_v$ (two linear programs per voxel) certify the query when the voxels with $L_v \geq \tau$ already connect p, q, or those with $U_v \geq \tau$ do not. Otherwise mixed-integer programs with separator and flow encodings search for the two witnesses, and a joint verifier bounds $\min b$ and $\max b$ exactly. These encodings are standard (formulations and validation in the supplement). A certificate covers all volumes allowed by the stated noise and object model; it is computed in floating point with independent re-checks, not a formal proof.

5. Experimental setup

All experiments are simulations with the optics of Sec. 3: 532 nm excitation and detection, excitation NA 0.8, detection NA 0.1, a phase plate with $0.32\,\mu$m features at $D = 6.6\,\mu$m, 256 directions, $0.08\,\mu$m optical and $0.32\,\mu$m analysis grids, $0.64\,\mu$m reconstruction voxels, linear fluorescence in a homogeneous non-scattering medium.

Full-size imaging. Reconstruction grid $16 \times 48 \times 48$ ($N = 36{,}864$, $30.7 \times 30.7 \times 10.2\,\mu\text{m}^3$), 12×12 camera samples per image, $M = 144k$, compression $C = N/M \in \{1, 4, 8\}$ for $k = 256, 64, 32$ (five random direction subsets for $C > 1$). Reference volumes are two-photon stacks of mouse pyramidal neurons from the public DEEP^2 data [29], min–max normalised and resampled to the analysis grid. Test set: two prespecified crops from each of six cells of three animals; the learned denoisers are trained on 459 crops of a fourth animal, preprocessed identically (blind Gaussian denoising, $\sigma \sim \mathcal{U}(0, 0.25)$). All solvers are tuned per encoder and compression on two vascular crops of a separate stack, on grids that extend automatically at their edges; no test volume is used for tuning or training.

Metrics. Besides PSNR we report foreground SSIM (3D SSIM averaged over the dilated mask of voxels above 10 % of the maximum) and the *local depth error*: the mean absolute difference between the true and reconstructed axial centroids of each lateral column, weighted by the true column brightness over columns above 10 % of the brightest. Two references calibrate it: a depth-blind oracle with the exact lateral structure and each column spread over the crop's mean axial profile scores $1.32\,\mu$m, and a flat estimate, uniform in depth, $2.15\,\mu$m.

Photon budget. All encoders share relay, pupil, detector and assembly. Their excitation is simulated on an optical grid larger than the field (an overfilled aperture, so beam edges lie outside it), and normalised to the same deposited dose. Counts are $\text{Poisson}(\kappa_0\, 256\, y_i/k)$: a fixed total budget shared by the k frames, with κ_0 set per specimen so that the random plate reaches 15 dB at $k = 256$ (1000 photons per pixel per frame at the mean signal). Plate and mask are evaluated in three prespecified realisations. The trained plate starts from the random plate (realisation 0) and changes only inside the specimen field.

Reduced structural protocol. Certification uses exact optimisation with dense rows, so it runs on a reduced field of the same optics: reconstruction grid $4 \times 12 \times 12$ ($N = 576$), 3×3 camera pixels, $M = 9k$, $k = 32$; the operator matches the live simulator to 3×10^{-7}. Queries use the central $4 \times 4 \times 4$ region, 6-adjacency, $\tau = 0.4$ and prespecified endpoints on 24 synthetic phantoms (filaments, bridges, bead chains, rings, dense volumes; supplement). Noise is Gaussian at 15 and 25 dB with three seeds; boxes are Bonferroni at $\alpha = 0.05$ ($z = 4.30$). Solves use HiGHS [9] (60 s, rerun at 600 s if undecided). Reduced-field plates are trained with the same objective for the reduced geometry. The reconstructions of Sec. 6.2 use box-constrained least squares with and without a TV term tuned on ten calibration phantoms; the regularity-prior and model-mismatch protocols are in the supplement.

6. Results

6.1. The encoder

Figure 3 and Tab. 1 compare the plate with three encoders that share its relay, detector, dose, objects, solvers and tuning protocol. As predicted (Sec. 3), the plate's excitation shears with tilt at 0.7–0.9 of the thin-screen rate, per-pixel excitation sequences decorrelate within about $1\,\mu$m of depth, and point emitters are assigned a wrong depth in 1.5–1.8 % of cases (three plates) against 4.4–5.3 % (three defocused masks), 25 % (DMD, half at the mirror depth) and 89 % without the plate. With every solver the plate and the mask reconstruct far better than the other two. The in-focus DMD stays at the depth error of a flat estimate (2.17 against $2.15\,\mu$m) while the plate reaches $0.36\,\mu$m (pretrained prior, $C = 1$). Without the plate the only depth signal leaks from the finite simulated aperture: enlarging it from two to four times the field cuts the angle-to-angle excitation modulation from 5.6 to 1 % (plate: 70 %) and raises the depth error from 0.38 to $0.75\,\mu$m ($C = 1$), towards the depth-blind oracle; Tab. 1 reports the larger aperture. The defocused amplitude mask

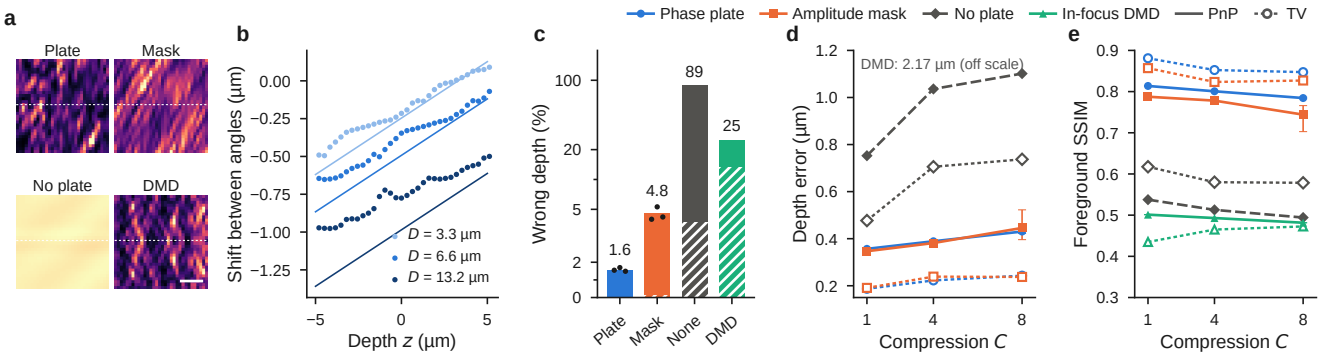


Figure 3. **Defocus under the angle scan encodes depth; in-focus patterns and a plain scan do not** (full-size field, equal deposited excitation dose, overfilled illumination aperture). (a) One excitation pattern per encoder, xz section through the specimen depth ($\pm 5\,\mu$m, focus dotted), each tile scaled to its own 99.5th percentile. (b) Lateral shift of the plate's excitation between two scan angles at every depth for plate defocus $D = 3.3$, 6.6 and 13.2 μm; lines: thin-screen prediction, sheared about the plate's conjugate plane at distance D. (c) Point emitters assigned a wrong depth by maximum likelihood with unknown brightness at $k = 32$ (dots: three plate and three mask realisations; hatched: mirror depth). (d, e) Reconstructions of 12 neuron crops from 6 cells: local depth error and foreground SSIM versus compression, multi-slice PnP (solid, no plate dashed; plate and mask at $C = 4, 8$: mean and range over three realisations) and box-constrained TV (dashed); the DMD's depth error of 2.17 μm is off scale. No plate: simulated aperture four times the field in (a, d, e), twice in (c).

		multi-slice PnP			box-constrained TV			beads (PnP)	
C	encoder	PSNR	FG-SSIM	depth	PSNR	FG-SSIM	depth	depth	ghost
1	phase plate	35.73	0.814	0.357	38.15	0.881	0.187	0.00	0.033
	amp. mask	35.24	0.788	0.347	37.59	0.857	0.191	0.00	0.030
	no plate‡	30.87	0.538	0.753	32.78	0.617	0.477	0.32	0.363
	in-focus DMD	28.96	0.501	2.168	28.48	0.435	2.285	3.84	1.007
4	phase plate†	35.40	0.801	0.389	37.34	0.852	0.223	0.01	0.013
	amp. mask†	34.88	0.778	0.382	36.79	0.823	0.239	0.01	0.017
	no plate‡	30.41	0.513	1.036	32.09	0.580	0.706	0.34	0.452
	in-focus DMD	28.91	0.493	2.168	28.49	0.465	2.326	2.94	1.013
8	phase plate†	34.86	0.784	0.430	37.00	0.847	0.243	0.01	0.039
	amp. mask†	34.14	0.744	0.446	36.74	0.827	0.237	0.08	0.048
	no plate‡	30.18	0.494	1.102	31.81	0.579	0.737	0.49	0.335
	in-focus DMD	28.83	0.482	2.169	28.53	0.473	2.311	2.60	1.013

Table 1. **Encoder comparison** (mean over 12 test crops; depth errors in μm; beads: depth error of detected beads and mirror-ghost energy ratio). †PnP: mean over three plate or mask realisations; TV and all other rows: realisation 0. ‡Simulated aperture four times the field (two times for all other rows; the plate changes by ≤ 0.07 dB). Equal deposited excitation dose; Poisson noise at a 15 dB operating point for the plate. Depth-blind oracle 1.32 μm, flat estimate 2.15 μm (Sec. 5).

setting (reduced field, $\tau = 0.4$)	certified	ambiguous	other
$k = 32$, 15 dB (72 queries)	24	48	0
trained plate (seeds 0 / 1)	33 / 29	39 / 43	0
$k = 256$, same photons	24	48	0
$k = 256$, $8\times$ photons	30	42	0
$k = 32$, 25 dB (72 queries)	60	12	0
trained plate (seeds 0 / 1)	63 / 63	9 / 9	0
TV budget, truth inside (18 runs)	0	18	0
TV budget, truth outside (9 runs)	0	4	5 empty
mismatch: exact coarse model (16)	0	1	15 empty
mismatch: noise only (32)	0	4	28 empty
mismatch: noise + allowance (64)	1*	63	0

Table 2. **Structural outcomes.** Top: 24 phantoms $\times$ 3 noise seeds, random plate unless stated. Middle: the nine ambiguous cases of Fig. 4 and the supplement under a calibrated total-variation budget $R_{\mathrm{cal}}, 2R_{\mathrm{cal}}, 4R_{\mathrm{cal}}$. Bottom: eight held-out phantoms measured on the fine grid, under two sub-voxel models. *This certificate contradicts the truth; the calibrated allowance failed to cover that object. Wrong certificates otherwise: 0.

comes close to the plate: over three realisations of each, the plate leads in PSNR in all nine pairings ($+0.4$ to $+1.2$ dB at $C = 8$) but not in depth error, so most of the depth encoding comes from defocus under the angle scan. Defocus, tuning, convergence and aperture checks are in the supplement.

6.2. What the measurements leave open

A reconstruction can show the true structure and still not be determined by the data. Figure 4 contrasts what a reconstruction shows with what the measurements support, on 13 reduced problems (nine proven ambiguous, four certified controls; all in the supplement). Every reconstruction (two estimators) lies inside $\mathcal{C}(\mathbf{y})$, so it is one consistent volume chosen by a prior, like the witnesses. Reconstructions reproduce the true topology in 11 of 13 cases, often with a clear margin (P01: bottleneck 0.64 against $\tau = 0.4$), yet in nine of the 13 the same images are explained by a volume of the opposite topology. In the dense volume P24 the reconstruction reports a disconnection that is false, and it fits the data better than the truth does (largest normalised residual 0.24 against 0.83). The opposite-topology witnesses are not rough artefacts: for filament phantoms their total

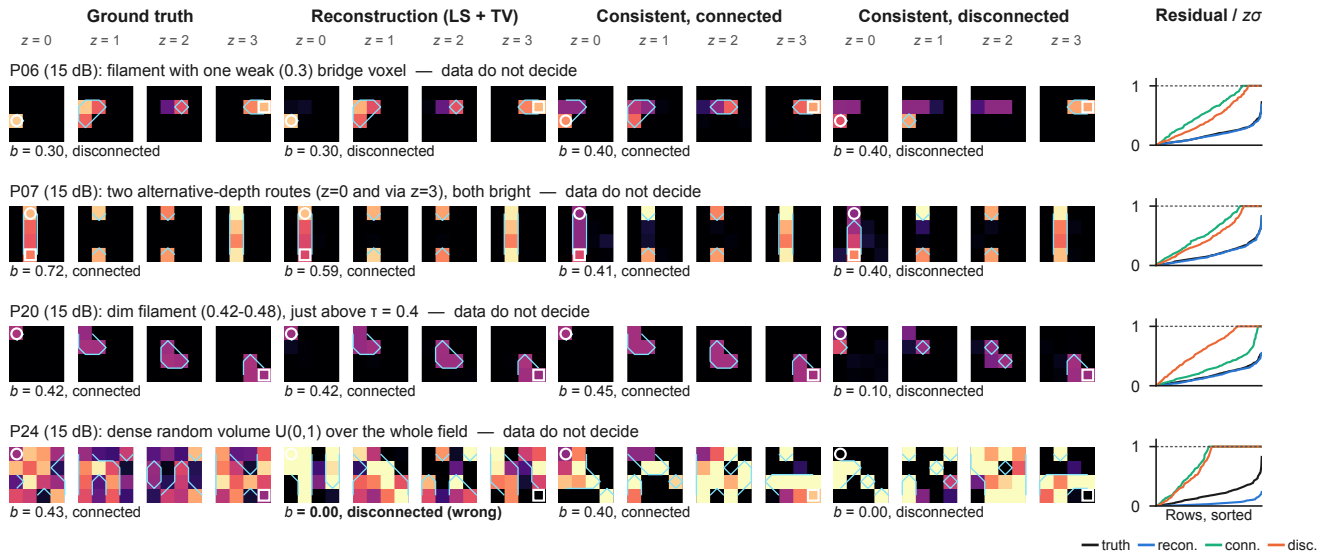


Figure 4. **One set of measurements, opposite topologies** (reduced field, 32 directions, 15 dB). Each row shows the $4 \times 4 \times 4$ query region as four depth slices: ground truth; box-constrained least squares + TV reconstructed from the recorded measurements; a volume in $\mathcal{C}(\mathbf{y})$ that connects p (circle) to q (square); a volume in $\mathcal{C}(\mathbf{y})$ that does not. Cyan outlines: voxels above $\tau = 0.4$; b: bottleneck. Right: sorted residuals $|(\mathbf{A}\mathbf{x} - \mathbf{y})_i|/z\sigma$; all curves stay at or below 1, so all four volumes explain the same images within the 95% box. The reconstruction is correct in the first three rows and wrong in the last; in all four the data do not decide the query.

variation is within 15 % of the truth's, and they mostly move brightness between depth-adjacent voxels, the freedom the plate objective targets.

Photons, not angles. At 15 dB, 48 of 72 queries are ambiguous with the random plate at $k = 32$ (Tab. 2). Spreading the same photons over all 256 directions resolves none of them and does not shrink per-voxel uncertainty; eight times the photons resolve 6. The trained plate resolves 9 at no photon cost (Sec. 6.3). Without noise, the operator certifies every query once it is square ($k = 64$); its condition number of 8.9×10^5 is why this does not survive noise.

A better prior does not decide. A calibrated regularity prior, restricting $\mathcal{C}(\mathbf{y})$ to volumes no rougher than the roughest of ten calibration phantoms (total variation $\leq R_{\text{cal}}$, also $2R_{\text{cal}}$ and $4R_{\text{cal}}$), leaves every query ambiguous whenever the truth satisfies the budget (18 of 18 runs): witnesses of the opposite topology exist that are as regular as the truth. When the budget excludes the truth, the set becomes empty or stays ambiguous; no run produced a certificate. Sub-voxel model mismatch makes the sets wider still, and a certificate is only as reliable as the error model (one wrong certificate in 64 runs, where the calibrated allowance missed the object; supplement).

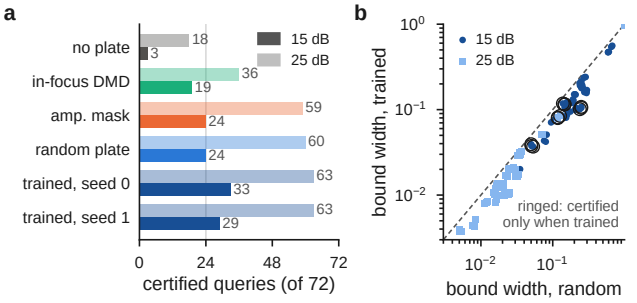


Figure 5. **The trained plate enlarges what the measurements determine** (reduced field, $k = 32$, 24 phantoms $\times$ 3 noise seeds). (a) Connectivity queries certified at 15 and 25 dB for every encoder; two training seeds of the plate. (b) Mean per-voxel bound width of each query, trained plate (seed 0) against the random plate it started from; ringed: queries certified only with the trained plate. No query certified with the random plate is lost, and no certificate is wrong.

6.3. Optics decide

Training changes which volumes the measurements allow (Fig. 5). On the reduced field the trained plate certifies 33 of 72 queries at 15 dB against 24 for the random plate it started from, and 63 against 60 at 25 dB; a second training seed gives 29 and 63. Paired query by query it gains 9 certificates at 15 dB and loses none, and no certificate is wrong. Per-voxel bounds narrow to $0.79\times$; the two-column rule of Sec. 4 predicts $0.80\times$ from the drop in mean depth

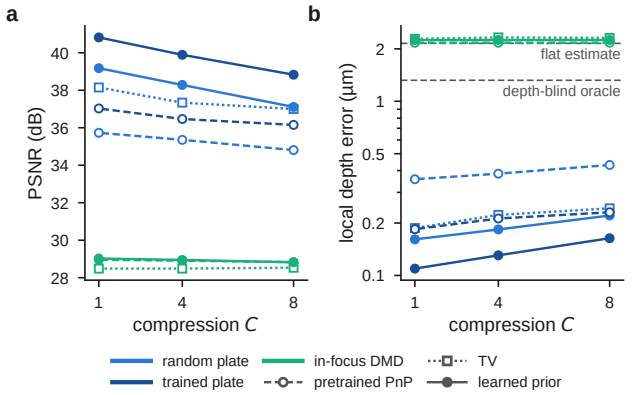


Figure 6. **Optics and prior gains add, where the optics record depth** (full-size field, 12 test crops, equal deposited dose). PSNR (a) and local depth error (b) against compression C; colour: encoder, line: solver (TV shown for the random plate and the DMD). Dashed lines in (b): a depth-blind oracle and a flat estimate. Training the plate and learning the prior each improve the plate; no solver moves the DMD off the flat estimate.

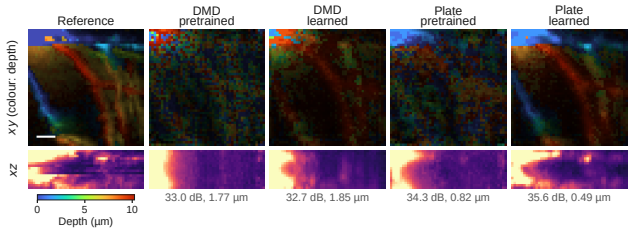


Figure 7. **A better prior, not a better answer** ($C = 8$, prespecified display crop, draw 0, identical display). Top: lateral maximum-intensity projection, colour coding the depth of the brightest voxel; bottom: xz projection; below: PSNR and local depth error. The learned prior sharpens the DMD's lateral structure but cannot give it depth; with the plate it recovers the depth layering. Bar: 5 μm.

correlation $|\rho|$ from 0.868 to 0.761 (across encoders it orders the widths correctly but overstates the differences where $\rho \to 1$; supplement). The gain survives fabrication: rounding the trained phase to 8 or 4 levels, or adding 0.5 rad of random phase error to the 8-level plate, keeps 32–33 certified queries with none lost.

At full size the same objective, trained once for the full field, halves the depth error with the pretrained prior at every compression (0.18 against 0.36–0.37 μm at $C = 1$, 0.23 against 0.42–0.44 μm at $C = 8$; three random plates tuned on the same grid) and adds 1.2–1.4 dB. Against each random plate its depth error is lower on 12 of 12 crops at every C and its PSNR on 10–12 of 12; a second training seed replicates this (PSNR on 9–12 of 12). With TV it lowers the depth error by 25–37 % (0.15 against 0.24 μm at $C = 8$, both seeds lower on 11–12 of 12 crops at every C) and with the learned prior by 26–32 % (0.16 against 0.22 μm at $C = 8$, lower on 11 of 12 crops), so the trained plate at $C = 8$ matches the depth error of the random plate with all 256 frames (Fig. 6).

6.4. Priors choose

Replacing the pretrained natural-image denoiser by denoisers trained on a held-out animal changes the volume chosen from the same measurements (Fig. 6). With the random plate the learned prior gains 2.3–3.5 dB and halves the depth error at every compression (39.2 dB, 0.16 μm at $C = 1$; 37.1 dB, 0.22 μm at $C = 8$), on 12 of 12 crops in both measures. It also beats the tuned TV solver in depth (10 of 12 crops at $C = 8$), though not consistently in PSNR; adding a TV term to it does not help on test (supplement).

The same prior cannot create depth information that the optics did not record. With the in-focus DMD every solver, learned or not, stays at 2.17–2.33 μm, the depth error of a flat estimate, and a second DMD pattern set reproduces every solver to within 0.06 dB and 0.07 μm (supplement). Splitting each column's error into its mean over depth (lateral error) and the zero-mean remainder (depth error) shows why. At $C = 8$ the learned prior shrinks the plate's lateral error to 0.48× and its depth error to 0.71× (median over crops; depth error lower on 12 of 12 crops), and the mask's to 0.37× and 0.55×. For the DMD it shrinks the lateral error to 0.77× but leaves the depth error at 1.04× (lower on 4 of 12 crops), and depth error is 99 % of the DMD's total. Figure 7 shows the consequence: with the learned prior the DMD's dendrites look sharper in xy, and its xz section is still a depth-less smear. The mask follows the plate (36.7 dB, 0.24 μm at $C = 8$); without the plate the learned prior improves depth error only from 1.10 to 0.92 μm ($C = 8$).

Forward-model accuracy. In a separate modelling comparison, simulating the optics directly at the reconstruction spacing gives a forward discrepancy of 25.7%, compared with 1.07% for the voxel-integrated operator. Voxel integration also considerably improves depth accuracy in the held-out reconstruction comparison (supplement).

7. Discussion and limitations

Division of labour. The two learned components did different jobs. Training the plate changed which volumes the measurements allow: more queries certified, narrower bounds, half the depth error. Training the prior changed which allowed volume was returned: better with the plate, unchanged in what is certified (by construction) and in depth for an encoder that records little. End-to-end training scores both with one reconstruction loss and cannot tell them apart; the consistency set can.

What to report with a reconstruction. At the operating SNR many connections a reconstruction shows are not determined by the data, and more directions at the same photon budget do not help. A connection reported from such a system should come with the outcome of the query. When one particular connection matters, the levers are the optics and the photon budget, not angular sampling or a stronger prior. Detection is part of the optics: at NA 0.3, 16 frames beat 256 at NA 0.1 (supplement).

Limitations. All results are simulations, without hardware validation, of linear fluorescence in a homogeneous non-scattering medium with an ideal relay and shot noise only. The test set has 12 crops from six cells of three animals, which are not independent; the plate is trained with two seeds, the prior on one animal, and the DMD has two pattern sets. The overfilled aperture was introduced after the first comparison revealed an aperture confound; both are reported. Certification runs on a $4 \times 12 \times 12$ field with oracle endpoints and a fixed threshold, never jointly with full-size reconstruction; certificates can be wrong when the model does not cover the object (one mismatch run), and their coverage is empirical.

8. Conclusion

In a computational microscope the optics decide which volumes the measurements allow; the prior only chooses one. On an angle-scanned phase encoder for 3D fluorescence, a plate trained with a data-free depth-decorrelation loss enlarges the set of connections the data certify, a learned prior improves the chosen volume but not what is certified, and photons, not angles, decide the rest. Learned optics should be judged by what their measurements determine, not only by how good the reconstruction looks.

LLM Usage Statement

LLMs were used for grammar and fact checking.

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